

Neurodevelopmental effects in children associated with exposure to organophosphate pesticides: A systematic review



Many studies have investigated the neurodevelopmental effects of prenatal and early childhood exposures to organophosphate (OP) pesticides among children, but they have not been collectively evaluated. The aim of the present article is to synthesize reported evidence over the last decade on OP exposure and neurodevelopmental effects in children. The Data Sources were PubMed, Web of Science, EBSCO, SciVerse Scopus, SpringerLink, SciELO and DOAJ. The eligibility criteria considered were studies assessing exposure to OP pesticides and neurodevelopmental effects in children from birth to 18 years of age, published between 2002 and 2012 in English or Spanish. Twenty-seven articles met the eligibility criteria. Studies were rated for evidential consideration as high, intermediate, or low based upon the study design, number of participants, exposure measurement, and neurodevelopmental measures. All but one of the 27 studies evaluated showed some negative effects of pesticides on neurobehavioral development. A positive dose–response relationship between OP exposure and neurodevelopmental outcomes was found in all but one of the 12 studies that assessed dose–response. In the ten longitudinal studies that assessed prenatal exposure to OPs, cognitive deficits (related to working memory) were found in children at age 7 years, behavioral deficits (related to attention) seen mainly in toddlers, and motor deficits (abnormal reflexes) seen mainly in neonates. No meta-analysis was possible due to different measurements of exposure assessment and outcomes. Eleven studies (all longitudinal) were rated high, 14 studies were rated intermediate, and two studies were rated low. Evidence of neurological deficits associated with exposure to OP pesticides in children is growing. The studies reviewed collectively support the hypothesis that exposure to OP pesticides induces neurotoxic effects. Further research is needed to understand effects associated with exposure in critical windows of development.